

[DC] Exploration of Context and Physiological Cues for Personalized Emotion-Adaptive Virtual Reality

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ABSTRACT

Immersive Virtual Reality (VR) can create compelling context-specific emotional experiences, but very few studies explore the importance of emotion-relevant contextual cues in VR. In this thesis, I investigate how to use combined contextual and physiological cues to improve emotion recognition in VR and enhance shared VR experiences. The main novelty is the creation of the first Personalized Real-time Emotion-Adaptive Context-Aware VR (PerAffectly VR) system which provides significant insight into how to create, measure, and share emotional VR experiences.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Interaction design—Interaction design process and methods—Contextual design

1 INTRODUCTION

My PhD thesis explores how to create personalized emotion adaptive VR experiences. Emotions are an essential part of our everyday lives, and play a critical role in social interactions, intra-personal experiences, and increasingly human-technology interaction. Emotions are affected by factors like social interactions, and past experiences encoded in and retrieved from Autobiographical Memory (AM). Communicating affective states with another person is an essential component for creating shared emotional experiences [9]. Ineffective affective communication produces a lack of empathy, which is a pressing issue among people, especially for those with depression. Emotionally responsive interfaces that respond to a person's current emotional state could be one way to improve people's quality of life.

Past researchers have explored multi-modal emotion recognition in VR, but there has been little research on how this could be combined with AM recall and other contextual cues to objectively measure emotional states, and using this to adapt the VR environments to provide a better user experience. This thesis explores how low-cost sensors can capture physiological signals (EEG, EDA, Heart Rate, etc), and combine them with contextual cues such as user's past experience, and current environment, to measure emotion. The main contribution is the design, development, evaluation, and demonstration of the first Personalized Real-time Emotion-Adaptive Context-Aware VR (PerAffectly VR) system. The research could be useful in multiple domains such as advancing wellbeing, tourism, training, entertainment, and gaming.

1.1 Research Questions

The main research question is "How to use the fusion of contextual and physiological cues to improve the recognition of emotions for personalized context-aware emotion-adaptive VR?". To investigate this primary research question, I am addressing the following sub research questions: (1) How to objectively measure psychophysiological responses associated with emotions in VR? (2) How

to develop a reliable and improved emotion-recognition model by providing contextual and physiological cues in VR?, and (3) How can we design a personalized real-time emotion-adaptive context-aware VR (Per-Affectly VR) system?

2 BACKGROUND

There has been a lot of research investigating emotion recognition techniques using various physiological modalities such as face expression, voice, text, and Eye-tracking [8]. VR headsets are now appearing with physiological sensors embedded in them to measure facial affective signals, and bring users' affective information into VR. For example, the HP Omnicept VR HMD¹ supports pupilometry and heart rate, while the upcoming Galea HMD² adds EEG and EMG sensing.

Early investigations of memory processes in immersive VR environments (VRE) reported enhanced presence, and highly vivid, emotional autobiographical memory recall in VR compared to conventional laboratory settings [1]. From these works, we can see that VR can possibly play a crucial role in emotional AM retrieval, yet researchers to date only relied on subjective questionnaires. Therefore, we aim to fill this gap by measuring EDA, HRV, and Pupil responses physiology during AM retrieval in VR.

Leveraging the personalized emotion-recognition methods, emotion-adaptive VR has not been explored extensively especially reflecting emotions back into the experience. Dey et. al. [3] reported positive influence of heart rate feedback to the collaborators on others' presence, and increased empathy. Bernard et. al. [2] demonstrated a VR system where user's avatar manipulate their expressiveness based on the user's real-time emotional states. These researches explored the emotions as an indicator but lacks the investigations on the impact of personalized emotion-adaptive system on user experience, and presence.

3 WORK TO DATE

To get familiar with the psychophysiological analysis, I conducted two VR experiments investigating psychological components like trust and cognitive load with the help of physiological signals (EEG, EDA, and HRV). The main hypothesis for the experiments was that the voice agents can assist people performing a VR and so reduce the person's cognitive load, but only if the agent is trusted. I published this in VRST 2019 [5], and IEEE VR 2020 [6].

I investigated emotion recognition techniques in VR system using EEG, and EDA information through AffectivelyVR, a personalized real-time emotion recognition system enabling an emotion-adaptive virtual environment. I used a machine learning based process to predict pleasant and unpleasant states using features extracted from EEG, and EDA signals, achieving an average accuracy of 96.5% for personalized models. I collected and labelled the data while participants' experienced 4 affective 360°VR videos and rated their emotional experiences using the Self-Assessment Manikin Questionnaire. For more, see [7].

¹ www.hp.com/us-en/vr/reverb-g2-vr-headset-omnicept-edition.html

² www.galea.co/

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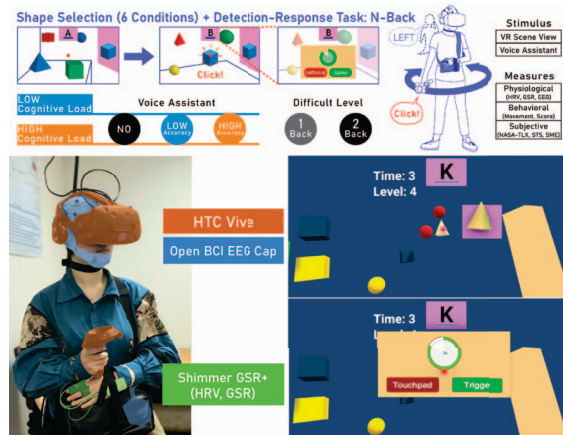


Figure 1: Experimental design, setup, and tasks for the psychophysiological study investigating relationship between cognitive load and trust in a Virtual Assistant in VR [6]

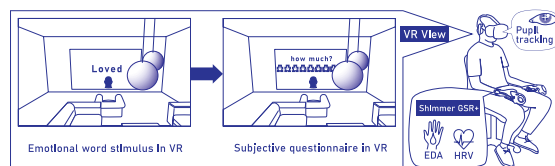


Figure 2: Experimental Setup and Design for AM recall using emotional words cues in VR

Next I explored if physiological sensing could be used to understand when AM were being recalled in VR. I replicated a conventional laboratory based Autobiographical Memory Test (AMT) where emotional text cues were used as stimulus to retrieve AM in VR. I displayed positive, negative, and neutral valence words to the participants and asked them to recall self-relevant memories associated with the words and collected EDA, HRV, Pupil Responses, and gaze. Results were presented at MobileHCI 2021 [4].

3.1 Future Work

I plan to explore the effects of immersive interactive virtual environments on emotional AM encoding and retrieval using physiological signals (EEG, EDA, HRV, and Pupil Responses) (RQ2). The primary focus is to provide insights on relationship of emotional AM recall with the physiological signals that can further be used as contextual information to improve the existing emotion recognition model. I will extend this study by combining environmental contextual cues such as user activity, VR location, and objects, along with the information about goals, and intention of the user with the existing emotion recognition techniques while they interact in the VR environment for further improvement.

Finally, for RQ3, I will develop PerAffectlyVR real-time emotion-recognition engine insights based on results from RQ1, and RQ2. This engine will be integrated with various Virtual Environments where it will predict the user's real-time emotional state and adapt the environment to optimize user's engagement, frustration, and overall experience. I will be testing the hypothesis: the PerAffectlyVR system provides a more immersive, personalized, engaging experience as compared to the non-adaptive VR system through interviews and questionnaires (usefulness, and usability).

4 EXPECTED CONTRIBUTIONS

The PhD research contributions are in two primary classes:

4.1 Artifact Contribution

(1) Custom VR environment to maximize or minimize cognitive load in the presence of a reliable or unreliable virtual agent; (2) A system using low-cost physiological sensors suitable for VR emotion capture; (3) PerAffectlyVR system, personalized real-time emotion-adaptive and context-aware VR system, that can adapt the VR environment, and difficulty level depending on the detected emotional states from the interactions.

4.2 Empirical Contribution

(1) Experimental design method in VR to evaluate the relationship between trust and cognitive load based on individual's cognitive load levels using physiological signals; (2) Methods to extract user's Contextual information such as AM recall, environment, and action in VR and integrate it with physiological signals to provide context-aware emotion recognition; (3) Findings from user studies conducted using PerAffectlyVR system to contribute to knowledge about personalized real-time context-aware emotion-adaptive VR technology.

5 EXPECTED BENEFITS OF PARTICIPATION

Advice from expert researchers in the IEEE VR community during the consortium will be invaluable and will greatly strengthen my dissertation. I believe such a consortium of minds will help to forge connections and potentially form collaborations which could kick-start further research in the field. I also will contribute to the discussions with fellow consortium participants. I have a deep research interest and experience in HCI methodologies, and rapid prototyping. With these interests, I know that I will provide new and unique perspectives for others to develop and enhance their research.

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